



Set Up a Web App Using AWS and VS Code



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```
1 <!DOCTYPE web-app PUBLIC
2     "-//Sun Microsystems, Inc.//DTD Web Application 2.3//EN"
3     "http://java.sun.com/dtd/web-app_2_3.dtd" >
4
5 <web-app>
6     <display-name>Archetype Created Web Application</display-name>
7 </web-app>
8
```



Introducing Today's Project!

In this project, we would be laying the foundations of a CI/CD pipeline by creating a webapp from scratch by launching an EC2 instance and connecting to it via VSCode to generate the web app within.

Key tools and concepts

Services I used were VS code and EC2. Key concepts I learnt include SSH connections using an IDE, launching an instance and editing an index.jsp file and using key pairs.

Project reflection

Was also using the terminal to edit the configuration file.

This project took me approximately an hour and 30 minutes. The most challenging part was using the VSCode to ssh. It was most rewarding to get it to work and getting an ssh connection

This project is part one of a series of DevOps projects where I'm building a CI/CD pipeline! I'll be working on the next project tomorrow



Launching an EC2 instance

I started this project by launching an EC2 instance because I would need a virtual server to host and manage my webapp

I also enabled SSH

SSH is a protocol that authorizes users to access remote servers. I enabled SSH so that it would allow the transfer of data between instances and to securely connect to our instance later. set ssh ip to my ip for security

Key pairs

A key pair is an access key that ensures secure access to your instance

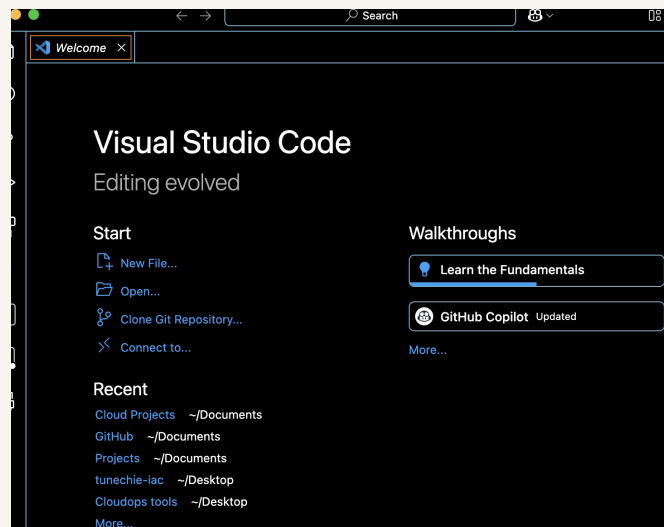
Once I set up my key pair, AWS automatically downloaded the private key file into my local machine



Set up VS Code

VS Code is a versatile tool for developers and programmers, offering a wide range of features and extensions to enhance code editing, debugging, and version control.

I installed VS Code to edit my webapp configuration files





My first terminal commands

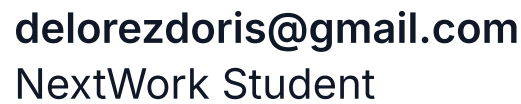
A terminal is where you send instructions to your computer to be executed. The first command I ran for this project is `cd ~/Desktop/DevOps`

I also updated my private key's permissions by running `chmod 400 + key pair name`

```
~/Desktop/Devops on ʘ main on ☁ (us-east-1)
> chmod 400 nextwork-keypair.pem

~/Desktop/Devops on ʘ main on ☁ (us-east-1)
> ls -l
total 8
-r-----@ 1 emehdoris  staff  1678 Apr 12 13:12 nextwork-keypair.pem

~/Desktop/Devops on ʘ main on ☁ (us-east-1)
> whoami
emehdoris
```



This command required an IPv4 address

```
/Desktop/Devops on 🐚 main on ☁️ (us-east-1)
➤ ssh -i /Users/emehdoris/Desktop/Devops/nextwork-keypair.pem ec2-user@3.89.104.24
```

The terminal output shows the following sequence:

- A decorative ASCII art logo consisting of wavy lines forming a stylized shape.
- The text "Amazon Linux 2023".
- The URL "https://aws.amazon.com/linux/amazon-linux-2023".
- The shell prompt "ec2-user@ip-172-31-30-80 ~\$".



Maven & Java

Apache Maven is a tool that helps developers build and organize Java software projects. It's also a package manager.

Maven is required in this project because it's really useful for kick-starting web projects! It uses something called archetypes, which are like templates, to lay out the foundations for different types of projects e.g. web apps.

Java is a popular programming language used to build different types of applications, from mobile apps to large enterprise systems.

Java is required in this project because if we don't install Java, we won't be able to use Maven to generate/build our web app today.



Create the Application

I generated a Java web app using the command `mvn archetype:generate`

I installed Remote - SSH, which is an extension within vs-code. I installed it to connect to my ec2 instance

Configuration details required to set up a remote connection include the host name i.e the instance ip, identity file -path to private key and the user for the instance

```
[INFO] -----
[INFO] Parameter: basedir, Value: /home/ec2-user
[INFO] Parameter: package, Value: com.nextwork.app
[INFO] Parameter: groupId, Value: com.nextwork.app
[INFO] Parameter: artifactId, Value: nextwork-web-project
[INFO] Parameter: packageName, Value: com.nextwork.app
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/ec2-user/nextwork-web-project
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 10.094 s
[INFO] Finished at: 2025-04-12T18:07:34Z
[INFO] Final Memory: 19M/85M
[INFO] -----
```



Create the Application

Using VS Code's file explorer, I could see source code files which contained the webapp folder and the pom.xml file

Two of the project folders created by Maven are src and webapp, which are the web app's files e.g. HTML, CSS, JavaScript, and JSP files, and resources, which are the configuration files a web app might need

The screenshot shows the VS Code interface. On the left, the Explorer sidebar displays the project structure: `src/main/resources/webapp/WEB-INF/web.xml`. The main editor area shows the content of `web.xml` with the following XML code:

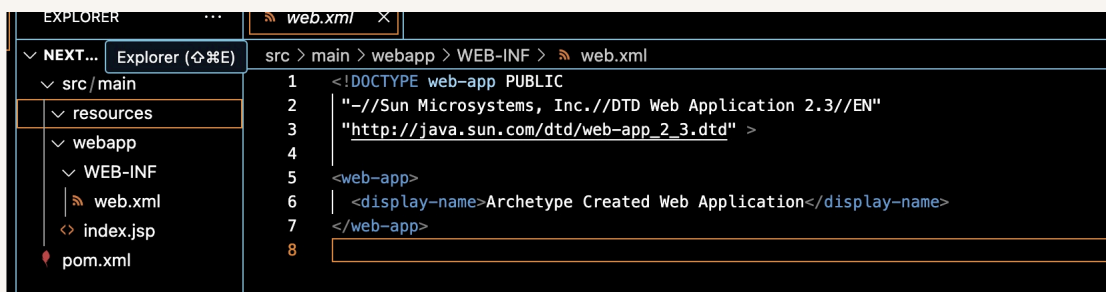
```
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2 "-//Sun Microsystems, Inc.//DTD Web Application 2.3//EN"
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Using Remote - SSH

index.jsp is a file used in Java web apps. It's similar to an HTML file because it contains markup to display web pages.

I edited index.jsp by updating the html code to say hello yourname and added a paragraph saying this is your web app working.



```
EXPLORER
  NEXT... Explorer (⇧⌘E)
  src > main
  resources
  webapp
  WEB-INF
    web.xml
    index.jsp
    pom.xml

web.xml
1 <!DOCTYPE web-app PUBLIC
2 "-//Sun Microsystems, Inc.//DTD Web Application 2.3//EN"
3 "http://java.sun.com/dtd/web-app_2_3.dtd" >
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